

Stein Fillings, Milnor Fibres, and Singularities: An Intuitive Approach

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June 27, 2026

1. ABSTRACT

The introduction portion serves as a bit of an "elevator pitch" for the rest of the paper. The general idea of the paper is about the relationship between Stein fillings and Milnor fibres using topological tools. This paper in particular explores the concept of Stein fillings, Milnor fibres, and singularities in an intuitive format. We aim to deduce why the Milnor fibre is worth exploring.

2. STEIN FILLINGS

Definition 2.1. A **stein manifold** is a complex manifold that admits a (proper) biholomorphic embedding in some $\mathbb{C}^n$.

Breaking this definition down, we describe a manifold to be a smooth space. So a complex manifold is a smooth (and complex) space. Manifolds are constructed by gluing smooth patches together, such that when we zoom in, they resemble a flat space. For example, the Earth is a manifold.

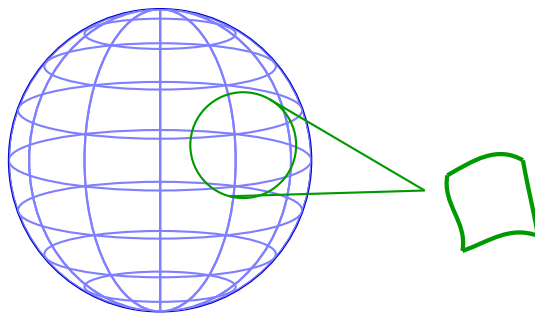


Figure 1: 3D manifold sphere with a local coordinate neighborhood patch.

Structures like a figure-8, or an "X" are not manifolds because the pieces don't glue together seamlessly. If one had a wooden structure of a figure-8 placed on a table and looked down on it, they would undoubtedly see that figure-8. But if they crouched down to see the structure at eye-level, they would see that the structure has junctions, which breaks the otherwise seamless structure.

In order for coordinate patches to glue together smoothly, the transition functions (or glue in this case) between the patches must be holomorphic.

Definition 2.2. The class of **holomorphic functions** is formed by the complex variable which possess a derivative wherever the function is defined.

In other words, to show that something is holomorphic, the aim is to show that a function is complex-differentiable (meaning it has a well-defined derivative and satisfies the definition of a limit). To achieve this, we employ the common method of using the Cauchy Riemann equations.

If $f(x, y) = u(x, y) + iv(x, y)$, then

$$\begin{aligned} f'(z) &= \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ &= -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \end{aligned}$$

Therefore,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \quad (1)$$

Eq. (1) are the Cauchy-Riemann equations. It's important to note that these equations don't tell us that something is smooth. In real geometry, this tends to be the case since we can deform shapes easily.

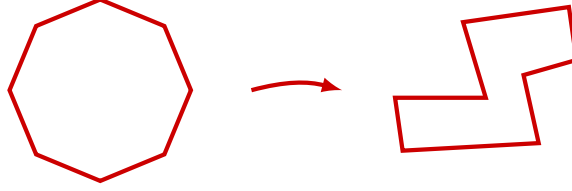


Figure 2: Regular vs irregular octagon.

Recall what it means for something to be **biholomorphic**? This means that we need to show that both a function f and its inverse f^{-1} are holomorphic.

Intuitively, you might think we could take the inverse of f and repeat the partial differentiations required to satisfy the Cauchy-Riemann equations. However, you may quickly realize that this is an extremely tedious process, and such methods of exhaustion only work for finite or simplistic cases. A shorter method is to use the **inverse function theorem (IFT)**.

Theorem 2.3. *The steps for IFT are as follows: A function $f(z)$ mapping a region U to a region V is biholomorphic if the following conditions are met.*

1. *It is bijective, meaning it is one-to-one and onto. Essentially, we are showing that different inputs yield different outputs. For something to be onto, we show that $\forall y \in Y, \exists x \in X | f(x) = y$.*
2. *Holomorphic on U and $f'(z) \neq 0$.*

This theorem ensures the structure is rigid in the complex space and isn't changing or evolving over time. What is meant by **proper embedding** is that a map $f : X \rightarrow Y$ between two topological spaces is proper if the pre-image of every compact subset in the target space is compact.

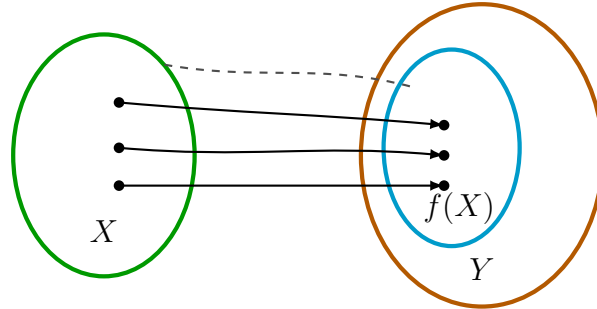


Figure 3: A topological or geometric proper embedding of space X into Y .

The idea behind a proper embedding is to prevent points from escaping to infinity such that the topology breaks. Sequences heading to infinity in the domain must do the same in the co-domain. We are saying the topological shape must be the same on the inside and outside.

An extremely important feature of the Stein manifold is that it can **never be compact**. Hence, we explore **Stein fillings**. Similar to the introductory concept of integration in calculus where we slice some area under a curve into little Riemann segments, a *filling* is a slice of the manifold.

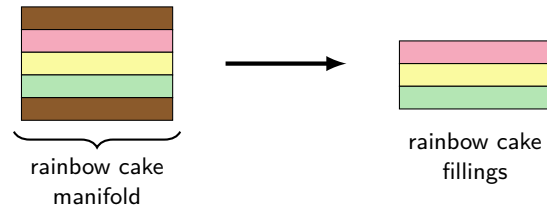


Figure 4: Conceptual breakdown of a stratified space using a layered cake analogy.

3. MILNOR FIBRES

Recall the concept of a Stein filling. Fillings can be thought of using a balloon analogy.

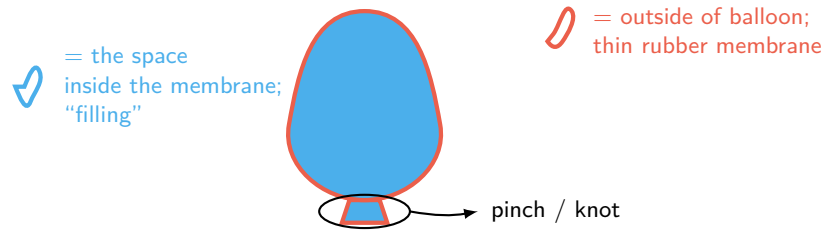


Figure 5: A geometric analogy explaining stratified fillings using a balloon structure.

Stein fillings are more or less a broad category that encompasses how one can fill this balloon. However, a *milnor fibre* is a concrete physical object that fills the balloon, and is built by algebraic equations.

One question of interest is how exactly do we get the milnor fibre inside the balloon, or how do we reach this concept. From the balloon, we notice that near the bottom, there is a "pinch." This is known as an **algebraic singularity**.

Now, say we take a string, and we tie it around the balloon in such a way that we separate the balloon and singularity from each other. One half is a sphere, and the other is this "pinch," which clearly has some broken geometry.

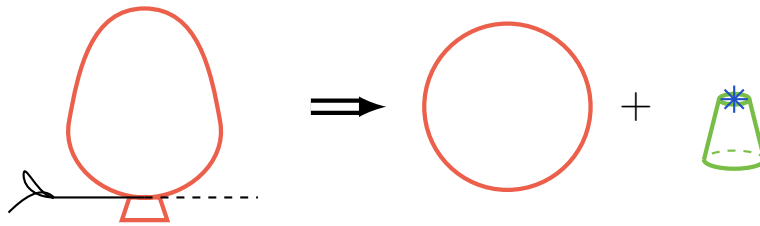


Figure 6: Cutting a balloon neck into a closed sphere and a boundary cylinder.

Notice that on the singularity, the tip is specifically what needs fixing. Right now, the "pinch" breaks the rigidity of the balloon such that the inverse theorem fails. In order to resolve this, we perturb the system. In *mathematics* perturbing a system does not mean provide it energy, but to remove the singularity. In essence, perturbation allows the structure to become rigid again!

Theorem 3.1. Milnor's fibration theorem states that for any small non-zero value of ϵ , the fibres are diffeomorphic; any choice of ϵ doesn't alter the shape of the Milnor fibre.

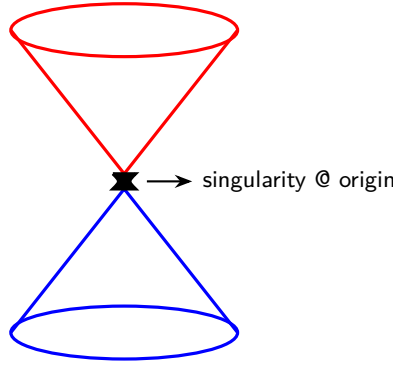
Now, say we that after perturbation, we put the string and singularity back together. So it is whole again. Since the singularity has been perturbed by a small amount ϵ , any change (geometrically) is trapped inside.

As we move from the centre towards the rubber membrane of the balloon, the effect of the perturbation practically fades to nothing. This satisfies what we know as a proper embedding! The only difference between this balloon and a normal balloon is that the pinch is wrinkle-free.

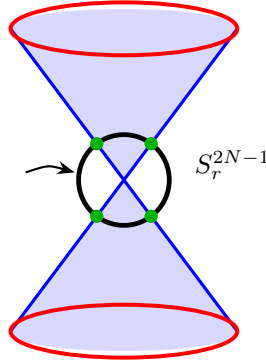
Therefore, the most important result is that **Milnor fibres are a category of Stein fillings.**

4. INTRODUCTION

The introduction begins by describing the shape in question. X is described as *some* shape that exists in $\mathbb{C}^n$. Intuitively, an isolated singularity at the origin sounds like an hour-glass shape, or from Einstein's special theory of relativity, a "light cone."



r is defined to be a Milnor radius. The intersection $Y = X \cap S_r^{2N-1}$ with a sphere $S_r^{2N-1} = \{|z_1|^2 + |z_2|^2 + \dots + |z_N|^2 = r\}$ is a 3D manifold called the *link of singularity* $(X, 0)$.



From the diagram above, the blue hour glass is X . The pinch at the centre is the singularity at the origin. The green dots denote Y . Notice that the "cuts" are far from the origin, so they appear to be smooth.

4.1. The Concept of S_r^{2N-1}

This notation describes a sphere. Recall that a complex number consists of two parts:

$$z = x + iy,$$

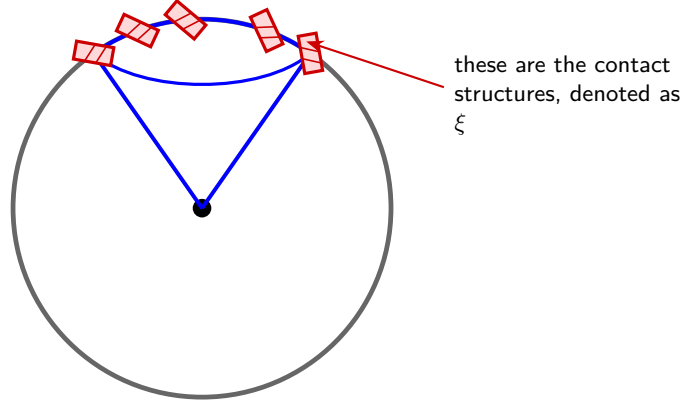
where the first term is a real term, and the second term is imaginary. Due to this notation, we see that one complex dimension equals two real dimensions.

Recall that our background space is $\mathbb{C}^n$. Counting them as real dimensions, it is the equivalent of $2N$. This brings us to the discussion of the sphere. If we think of a spherical object, such as a ball, it consists of a filling (air) and a membrane of some distinct material. If we look closely at this membrane, we realise that it is 2D, and simply bent into a rounded shape.

Mathematically, a sphere's surface always has one less dimension than the space it occupies. So in 3D, the sphere's surface is 2D, and in 2D, it's 1D.

5. CONTACT-

The contact structure can be described as a field of tiny twisty planes attached to "cuts" from Y . They twist only because they are in the complex space.



These induced contact structures ξ on Y is the distribution of tangencies to Y , and are referred to as the *canonical* or *Milnor fillable* contact structure on the link. (Y, ξ) is the *contact link*. It consists of the intersection on the sphere, as well as the tangential manifolds.

One of the most immediate questions proposed by the paper is why (Y, ξ) may be r -independent? (Y, ξ) is r -independent because changes in the radius only scales the size of the manifold without deforming the shape or twisting the induced contact structures.

Definition 5.1. A **contactomorphism** is a special kind of mapping. It's used to show that different contact manifolds are exactly the same. The conditions are:

1. It is homeomorphic
2. It has *twist-preservation*

So when r is changing, we get different (or unequal "cuts"). Contactomorphism proves that the radius doesn't deform the shape.

A problem in the topology of the surface singularity is to compare the Milnor fibres of smoothings $(X, 0)$ to symplectic (or contact) Stein fillings of (Y, ξ) . At this point, it's important that we adjust our definition of a Milnor fibre. This is the stuff that fills the singularity, and a Milnor fibration is the entire structure of fibres. Note again that Milnor fibres (and therefore fillings) are a category of Stein fillings.

Another important note is that *resolution* $\neq$ *perturbation*. A perturbation is something we know to smooth out a deformity. A resolution, however, can be seen as something that replaces the singularity. In sequence, we have a broken shape. From the minimal resolution, we look at the symplectic structure of the resolved shape and then deform it in such a way that we don't tear it. This makes a new Stein filling that agrees with a Milnor fibre of Artin smoothing component. At this point, we won't trouble ourselves mathematically.

Another question proposed by the paper is: does the collection of expected fillings, taken *forall* singularities with the same contact link (Y, ξ) , gives all possible Stein fillings of the link.

Definition 5.2. An **unexpected Stein filling** is any Stein filling that refers to Stein fillings that don't arise as a Milnor fibre or the minimal resolution.

The aim of this paper is to show Stein fillings are abundant and simply connected. Note that rational singularities with a reduced fundamental cycle are known as minimal singularities.

5.1. Minimal Singularities

In short, they are a type of singularity that are characterized by certain conditions.

- i) **The shape is a tree:** If you trace a path from vertex to vertex, you can never loop back to where you started. There are no closed cycles or loops in the graph.
- ii) Each single vertex represents a smooth projective line $\mathbb{P}^1$, which topologically is just a 2-dimensional sphere.
genus 0 = no holes.
- iii) **Valency** $[a(v)]$: # of edges attached to the vertex.
Self-intersection $(v \cdot v)$: Because the intersection matrix must be negative definite, this # must always be negative.
 $\hookrightarrow$ if you drop the negative sign from the self-intersection #, that # must be $\geq$ to the valency.

6. THEOREMS IN §1

Theorem 6.1. *For any $N > 0$, there is a rational singularity with reduced fundamental cycle whose contact link (Y_N, ξ_N) admits at least N pairwise non-homeomorphic simply-connected Stein fillings, none of which is diffeomorphic to a Milnor filling (rel certain boundary data). Examples of such (Y_N, ξ_N) include Seifert fibered spaces over S^2 corresponding to certain star-shaped resolution graphs.*

The statement also holds for any rational singularity with reduced fundamental cycle whose resolution graph has a star-shaped subgraph as above.

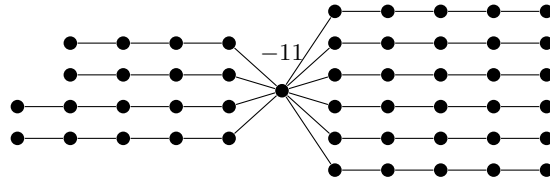
Recall the definition of the contact link (Y, ξ) . It consists of the intersection Y on the manifold and the tangential manifolds ξ . Let N be some number.

This theorem is saying that for any number $N > 0$, there is a rational singularity that satisfies the condition of a reduced fundamental cycle whose contact link (Y_N, ξ_N) admits N topologically distinct Stein fillings (*pairwise non-homeomorphic*).

Recall that diffeomorphic means you can reshape one object into another w/o tearing, folding, or making sharp corners. To achieve this, the following requirements must be met:

- i) *Smooth*: you can take its derivative (no sharp kinks/sudden jumps)
- ii) *Bijective*: every point on shape 1 matches to a point in shape 2.
- iii) *Invertible*: an inverse must exist, & it must be smooth.

These Stein fillings are structurally equivalent to Milnor fillings. Below is fig. 1 from the paper.

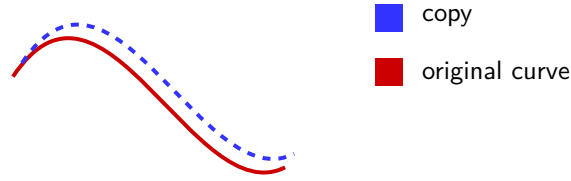


This is a **star-shaped resolution graph**.

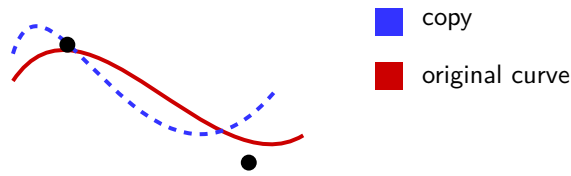
$$\begin{array}{c} -11 \\ \bullet \\ \underbrace{\hspace{1cm}} \end{array}$$
 this is the centre of the graph, which represents the 4D space.

The centre describes an S^2 , or 2-sphere. Recall again from the balloon analogy, this is a circular membrane that was wrapped around the bubble of air to create the 3-dimensional balloon shape.

The number -11 denotes the self-intersection number.



As you can tell from this diagram, we see that since the two curves don't intersect, the self-intersection number is 0.



Moving the copy line slightly, it begins to overlap in some way. This way is by a form of "twisting." We can't move the blue line back in such a way that it becomes parallel to the red line again. The intersections are denoted by the black dots.

Applying this concept to the star-shaped resolution graph, the copy around the central vertex, or balloon membrane is so intensely twisted such that when you try to make a copy of the balloon, it's physically forced to warp and pierce our "original" balloon 11 times.

So this number measures the severity of the twist. The negative sign implies the twist is pulling inward into a tight pinch.

Looking at the legs that branch out, they have black dots too. They are also balloons of S^2 . As labelled, each of these dots have a self intersection number, but of -2 . The lines between each dot are connections. We are forming a chain of balloons.

Theorem 6.2. *Let $(X, 0)$ be a rational singularity with reduced fundamental cycle with link (Y, ξ) , and suppose that each exceptional curve in its minimal resolution has self-intersection at most -5 . Then the resolution of $(X, 0)$ is the unique weak symplectic filling of (Y, ξ) , up to blow-up, symplectomorphism and symplectic deformation.*

Essentially, this theorem says that supposed we have a tangled singularity whose central value is -5 . The smooth geometric space we get from un-tangling the singularity naturally serves as a filling.

If you try to build another smooth compact 4-symplectic manifold that has (Y, ξ) as the boundary, it is guaranteed to be structurally identical to this "minimal resolution."

Definition 6.3. The term **up to blow up** means that we replace a point with a sphere that has a self-intersection of -1 . If another filling exists, the only way it can differ topologically is if it has these spheres glued to it in some way. If you take a pin, and you pop these spheres, you retain the minimally resolved figure.

Definition 6.4. A **symplectomorphism** means spaces are identical up to a smooth structure preserving map/isomorphism/diffeomorphism.

Definition 6.5. A **symplectic deformation** means that even if the exact angles or sizes of the symplectic shapes look slightly different on the surface, you can smoothly morph one symplectic structure into another without breaking the underlying geometric values.

6.1. The Conjecture of Kollár

Conjecture 6.6. Let X be a smooth complex projective variety. If X has a generically large fundamental group, then $\chi(X, K_X) \geq 0$.

This conjecture is a statement about the structured behaviours of deformations. It asserts that if a rational singularity has a minimal resolution where every exceptional curve has a self-intersection (of at most -5), then the base space of its semi-universal deformation consists of exactly one irreducible component (which is the *Artin* component).

6.2. Betti Numbers

Betti numbers are topological invariants that count the number of holes in a geometric space at different dimensions.

$$\begin{cases} \beta_0 \text{ (0D holes)} & : \text{Counts \# of connected components.} \\ \beta_1 \text{ (1D holes)} & : \text{Counts \# of circular tunnels/loops.} \\ \beta_2 \text{ (2D holes)} & : \text{Counts \# of hollow cavities/spheres.} \end{cases}$$



Solid Ball

$$\begin{aligned} \beta_0 &= 1 \\ \beta_1 &= 0 \\ \beta_2 &= 0 \end{aligned}$$



Circle

$$\begin{aligned} \beta_0 &= 1 \\ \beta_1 &= 1 \\ \beta_2 &= 0 \end{aligned}$$



Basketball

$$\begin{aligned} \beta_0 &= 1 \\ \beta_1 &= 0 \\ \beta_2 &= 1 \end{aligned}$$

Theorem 6.7. Let $(X, 0)$ be a rational singularity with reduced fundamental cycle, and $\mathcal{C} \subset \mathbb{C}^2$ its decorated germ of a reducible complex curve such that all the branches $C_1, \dots, C_m$ of $\mathcal{C}$ are smooth complex disks. Then smoothings of $(X, 0)$ are in one-to-one correspondence with picture deformations of $\mathcal{C}$. A picture deformation gives an arrangement $\mathcal{C}^s$ of the deformed branches $C_1^s, \dots, C_m^s$, $s \neq 0$, with marked points that include all the intersections of the branches. The Milnor fiber $W_{\mathcal{C}^s}$ of the corresponding smoothing can be constructed by blowing up at all marked points and taking the complement of the proper transforms of $C_1^s, \dots, C_m^s$.

The main concept in this theorem is the idea of a picture deformation. The picture deformation gives the branches a nudge such that the singularity breaks into "double intersection points."

Definition 6.8. A **blow-up** is a complex geometric operation. Points are taken and replaced with P^1 (or S^2). This eliminates the double intersections. By inserting S^2 at the points, the lines are forced to separate. They now pierce the sphere at different locations based on the angle at which they originally approached the point.

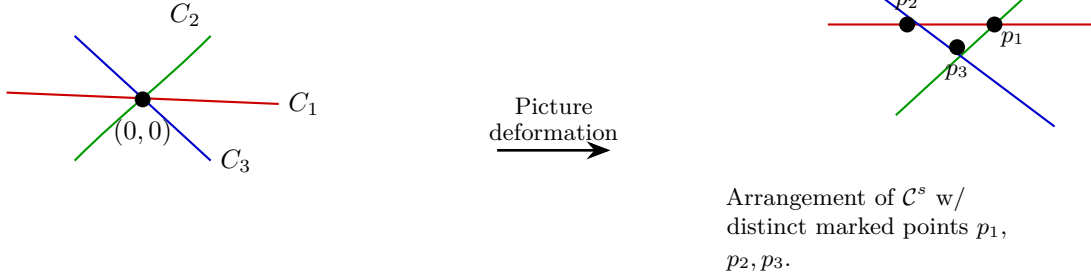


Figure 8: Deformation of a curve singularity into distinct transverse intersections.

Definition 6.9. When you blow up the space, old curves are pulled apart. **Proper transforms** are just the new name for the separated disk branches in the new/modified space. = disjoint smooth disks.

Definition 6.10. To take the **complement** is to take the entire 4D space and remove the disjointed disks. This leaves us with the Milnor fibre W_{C^s} .

The germ $\mathcal{C}$ is more formally identified as a curve. This curve and all of its branches exist within a ball B . Note that we only care about how the curve behaves arbitrarily close to the origin. Intersections of ∂B happen transversely.

To obtain a smooth compact 4D manifold such that the boundary link is Y :

- We blow up B at the marked points.
- Take the proper transforms and put it in that spot.

Definition 6.11. A **smooth graphical homotopy** essentially takes a complicated set of curves into a clear arrangement.

A homotopy is a way to deform one shape into another continuously; “smooth” implies that the morphing happens without any sudden breaks, sharp folding, or tearing. The importance of this is that it allows transverse and positive intersections.

A positive intersection is one that means the orientation of the crossing matches the standard orientation of the space.

Theorem 6.12. Let (Y, ξ) be the contact link of a singularity $(X, 0)$ with reduced fundamental cycle, and let $\mathcal{C}$ be a decorated plane curve germ representing $(X, 0)$, with m smooth components $C_1^0, \dots, C_m^0$. For any smooth graphical homotopy, let W be the smooth 4-manifold obtained by blowing up at all marked points and taking the complement of the proper transforms of $C_1^1, \dots, C_m^1$. (In the case of a picture deformation $\mathcal{C}^s$, W is the Milnor fiber W_{C^s} from Theorem 1.3).

Then W carries a planar Lefschetz fibration that supports a Stein filling of (Y, ξ) . When $W = W_{C^s}$, the Lefschetz fibration is compatible with the Stein structure on the Milnor fiber.

The fiber of the Lefschetz fibration on W is a disk with m holes, and the vanishing cycles can be computed directly from the decorated curve configuration $C_1^1, \dots, C_m^1$. On (Y, ξ) , the Lefschetz fibration induces a planar open book decomposition, which is independent of the smooth graphical homotopy of the given decorated germ $\mathcal{C}$.

This is describing the smooth manifold W . When W carries a planar Lefschetz fibration, it means it has some geometric nuance such that the 4D space can be broken into simpler 2D spaces/surfaces (fibres) that are completely flat in terms of their topology.

Remark 6.13. To describe this simply, let W be a loaf of bread. When we cut it into slices, each 2D slice is called a **fiber**. In a standard fibration, all the bread slices must be identical; however, the Lefschetz fibration allows some slices to have a specific type of defect.

These defects are that as we move across the base space:

- The 2D slice/fiber stays smooth, but certain parts start to shrink.

- At a few isolated critical points, these spaces shrink all the way down to a single point, causing the fiber to pinch tightly like an hourglass.

The surface is **planar** if it can be mapped entirely onto a flat sheet of paper:

$$\text{planar} = \text{genus } 0$$

This effectively shows what we do to make a 4D problem into a 2D problem.

Theorem 6.14. *Let (Y, ξ) be the contact link of a singularity $(X, 0)$ with reduced fundamental cycle that corresponds to a decorated plane curve germ $\mathcal{C}$. Then any Stein filling of (Y, ξ) arises from an arrangement Γ of symplectic graphical disks with marked points, as in Theorem 1.4. The arrangement Γ is related to the decorated germ $\mathcal{C}$ by a smooth graphical homotopy.*

This is like the research statement of how we are able to deduce all Stein fillings. For now, note that Γ represents the collection/configuration of 2-D geometrical disks that are found in the 4D-ball space.

7. EXTRA NOTES

7.1. Braided Wiring Diagrams

Monodromy factorization helps to describe how topological features, like singularities, branches, and fibrations can be broken down into simpler geometric components. In order to visualise this information, we use *braided wiring diagrams*.

Since we can't directly look at things in the 4-dimensional space $(\mathbb{C}^2)$, we take a slice of it to cut the dimensions down $(\mathbb{C} \times \mathbb{R})$. When the curve $\mathcal{C}^s$ passes through this 3-dimensional slice, their intersection comprises of 1-dimensional lines. These lines make up a braided wiring diagram.

7.2. Curvetas

A *curvetta* is described to be a small, flat 2-dimensional disk that pierces another surface at a singular point *transversely*. This particular intersection direction matters because it ensures stability at that point. If you perturb or wiggle a transverse intersection slightly, it stays an intersection. If you wiggle a tangent intersection, it can completely detach or split apart.

Going back to the idea of creating a chain of balloons, we had a collection of separate, smooth, individual complex disks $(C_1, C_2, \dots, C_m)$. Each one is an independent geometric object piercing its own separate surface.

If we collapse the background surfaces down to the origin $(0 \in \mathbb{C}^2)$. Because the surfaces vanish into that single dot, all the curvetas are dragged inward until they all pass through that exact same dot.

Once they are all forced through the origin, you can no longer treat them as completely isolated. You look at their union as a single geometric entity:

$$\mathcal{C} = C_1 \cup C_2 \cup \dots \cup C_m$$

This union $\mathcal{C}$ is a reducible curve germ because it is built from multiple distinct “branches” or “irreducible components” (C_i) that meet at a single highly singular point (the origin).

As a side note, In algebraic geometry, a curve is reducible if its defining algebraic equation can be factored into a product of simpler equations, such as:

$$f(x, y) = f_1 \cdot f_2 \cdot f_3 = 0$$

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